

Substrate Cognitive Dynamics: Hierarchical Neural Networks, Predictive Processing, Memory Integration, and Emergent Cognition

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Abstract

Building upon the biological process mechanics established in Paper X, we formulate a deterministic continuum framework for neural information processing, predictive processing, working memory, attention, decision making, and emergent cognition. In standard neuroscience, cognition is modeled as statistical rate-coding across discrete connectionist neurons. In the Unified Substrate Theory, we prove that the central nervous system is a multi-scale, coupled order-parameter phase network $\mathbf{N}_{\text{brain}}(\mathbf{x}, t) = \sum \mathbf{n}_i(\mathbf{x}, t) \in S^2$ operating as a global gradient optimizer. We derive the Predictive Error Minimization Theorem, demonstrating that internal generative models reduce sensory phase mismatch ($\Delta\Phi \neq 0$) by adjusting network streamtube topologies. We model attention as selective phase amplification ($A_0 + \Delta A$), formulate decision making as competitive energy relaxation across nested attractor basins, and re-frame collective neural oscillations (alpha, beta, gamma, theta) as global acoustic standing-wave modes.

1 From Single Neurons to Massively Coupled Phase Networks

In Paper X [1], individual axonal action potentials were derived as solitary substrate phase pulses $\mathbf{n}(z - v_{\text{nerve}}t)$ propagating along lipid-sheathed channels. In cognitive neuroscience, intelligence emerges from the collective interaction of over 10^{11} interconnected neural conduits.

In the Unified Substrate Theory (UST), the central nervous system is modeled as a continuous, multi-scale order-parameter phase field $\mathbf{N}_{\text{brain}}(\mathbf{x}, t) \in S^2$:

$$\mathbf{N}_{\text{brain}}(\mathbf{x}, t) = \sum_{i=1}^{N_{\text{neurons}}} \mathbf{n}_i(\mathbf{x}, t) e^{i\Phi_i(\mathbf{x}, t)}. \quad (1)$$

The network evolves via coupled order-parameter strain relaxation:

$$\frac{\partial \mathbf{N}_{\text{brain}}}{\partial t} = -D_{\text{cortex}} \nabla^2 \mathbf{N}_{\text{brain}} - \gamma \frac{\delta E_{\text{network}}}{\delta \mathbf{N}_{\text{brain}}} + \mathbf{S}_{\text{sensory}}(\mathbf{x}, t), \quad (2)$$

where $\mathbf{S}_{\text{sensory}}$ represents incoming order-parameter perturbations driven by peripheral sensory organs.

2 Predictive Error Minimization and Internal Generative Models

Rather than passively processing sensory inputs, the cortex actively generates top-down predictive phase patterns to anticipate environmental perturbations.

Theorem 1 (Predictive Error Minimization Theorem). *The total information-processing cost $E_{\text{prediction}}$ incurred by a cortical network is the spatial order-parameter strain differential between bottom-up sensory phase inputs $\nabla \mathbf{N}_{\text{sensory}}$ and top-down predictive model waves $\nabla \mathbf{N}_{\text{model}}$:*

$$E_{\text{prediction}} = \frac{1}{2} \mu_0 \int_{\Omega_{\text{cortex}}} |\nabla \mathbf{N}_{\text{sensory}}(\mathbf{x}, t) - \nabla \mathbf{N}_{\text{model}}(\mathbf{x}, t)|^2 d^3x. \quad (3)$$

Learning adjusts synaptic streamtube topology (Paper X) along steepest-descent trajectories to minimize $E_{\text{prediction}}$ ($\Delta E_{\text{prediction}} \rightarrow 0$).

Proof. When top-down predictive waves match incoming sensory phase fronts ($\mathbf{N}_{\text{model}} \approx \mathbf{N}_{\text{sensory}}$), destructive strain interference eliminates localized phase gradients ($\nabla \mathbf{N}_{\text{sensory}} - \nabla \mathbf{N}_{\text{model}} = \mathbf{0}$). The net strain energy functional $E_{\text{prediction}}$ drops to a local minimum, suppressing prediction error signals and conserving metabolic ATP energy (Paper X). An unexpected sensory input produces a phase mismatch ($\Delta \Phi \neq 0 \implies \nabla \delta \mathbf{N} \gg 0$), propagating error pulses upward through the cortical hierarchy to update internal phase topology. $\square$

3 Working Memory as Persistent Oscillatory Phase Loops

In Paper X, long-term memory was established as persistent structural phase-locking ($\Delta E_{\text{synapse}} < 0$). Working memory represents the short-term, active maintenance of information without permanent structural remodeling.

Proposition 1 (Working Memory Phase Loop Invariant). *Active working memory corresponds to a self-reinforcing, closed standing-wave phase loop traversing localized cortico-thalamic streamtube circuits. The active representation $M_{\text{active}}(t)$ is sustained by periodic phase recirculation:*

$$M_{\text{active}}(t) = \oint_{C_{\text{loop}}} \mathbf{N}_{\text{brain}}(\mathbf{x}, t) \cdot d\mathbf{l} \equiv \text{const}, \quad (4)$$

persisting until metabolic energy decay or external phase interference disrupts loop recirculation.

4 Attention as Selective Phase Amplification

Cognitive attention is the selective allocation of computational resources to specific sensory or internal channels.

Theorem 2 (Selective Phase Amplification). *Top-down attentional modulation enhances the local signal-to-noise ratio of a target neural channel Ω_{target} by injectively scaling its phase amplitude vector from baseline A_0 to $A_0 + \Delta A$:*

$$\mathbf{N}_{\text{attended}}(\mathbf{x}, t) = (A_0 + \Delta A) \mathbf{n}_{\text{base}}(\mathbf{x}, t) e^{i\Phi(\mathbf{x}, t)}. \quad (5)$$

Proof. Increasing local phase amplitude ΔA elevates the strain energy density $\mathcal{E}_{\text{target}} = \frac{1}{2} \mu_0 |\nabla \mathbf{N}_{\text{attended}}|^2$ relative to surrounding un-attended background channels ($\mathcal{E}_{\text{background}} \ll \mathcal{E}_{\text{target}}$). Attended phase fronts dominate global coupled network relaxation (Eq. 2), forcing downstream decision attractors to align with the amplified target signal. $\square$

5 Decision Formation as Competitive Energy Basin Relaxation

In classical decision theory, choice selection is modeled via stochastic drift-diffusion processes. In UST, decision making is deterministic relaxation across competing topological energy basins.

Theorem 3 (Competitive Attractor Decision Theorem). *Let K mutually exclusive behavioral choices correspond to K distinct local strain energy minima $\{E_1, E_2, \dots, E_K\}$ on the network state manifold $\mathcal{M}_{\text{cortex}}$. The network trajectory $\xi(t)$ evolves deterministically toward the deepest accessible attractor basin:*

$$\frac{d\xi(t)}{dt} = -\nabla_{\xi} E_{\text{network}}(\xi) + \mathbf{S}_{\text{sensory}}(t). \quad (6)$$

A commitment decision occurs when $\xi(t)$ crosses a unstable phase threshold $\xi^{\dagger}$, collapsing the network state into a single stable motor action basin.

6 Collective Neural Oscillations as Substrate Standing Waves

Electroencephalographic (EEG) frequency bands—delta (1–4 Hz), theta (4–8 Hz), alpha (8–12 Hz), beta (13–30 Hz), and gamma (30–80 Hz)—are standard signatures of cognitive state.

Theorem 4 (Substrate Neural Resonance Invariant). *Macroscopic brain rhythms are the global acoustic standing-wave eigenmodes of the coupled order-parameter field $\mathbf{N}_{\text{brain}}$ bounded by cortical cavity geometry Ω_{cortex} :*

$$\nabla^2(\delta\mathbf{N}) - \frac{1}{v_{\text{phase}}^2} \frac{\partial^2(\delta\mathbf{N})}{\partial t^2} = \mathbf{0} \implies \omega_k = v_{\text{phase}} \cdot k, \quad (7)$$

where v_{phase} is the effective intra-cortical shear conduction velocity (Paper VIII).

Proof. Gamma oscillations (30–80 Hz) represent localized, short-range standing waves mediating fine-grained feature integration ($\nabla_{\text{local}}\mathbf{N}$). Theta (4–8 Hz) and Alpha (8–12 Hz) rhythms correspond to long-range, cross-cortical standing waves coordinating large-scale phase synchronization between sensory areas and sub-cortical memory structures (Paper X). $\square$

7 The Complete Eleven-Paper UST Scale Progression

With Paper XI, the Unified Substrate Theory constructs a continuous, 11-paper deterministic continuum spanning sub-atomic physics to emergent cognition:

8 Conclusion

We have demonstrated that emergent cognition, predictive processing, working memory, attention, decision formation, and macro-scale EEG rhythms are direct consequences of coupled order-parameter phase-network optimization ($\min E_{\text{prediction}}$). By unifying physics, chemistry, biology, and cognitive science under a single continuum field framework, UST establishes that intelligence is a multi-scale, adaptive manifestation of substrate gradient-energy minimization.

References

- [1] R. Beem, *Substrate Systems Biology: Metabolic Strain Reservoirs, Molecular Machinery, Neural Phase Pulses, and Emergent Adaptive Dynamics*, UST Paper X (2026).

Table 1: **The Unified Substrate Theory (UST) Complete Scale Ladder**

Volume	Core Field Domain	Primary Governing Invariant
Paper I	Soliton Physics	Skyrme topological energy minimum
Paper II	Hadronic Physics	Topological proton rest mass (m_p)
Paper III	Electrodynamics	Fine-structure constant (α^{-1})
Paper IV	Gravitation	Pressure-shadow deficit (G)
Paper V	Atomic Chemistry	Orbital cavity eigenmodes & $2n^2$ shells
Paper VI	Organic Chemistry	Carbon σ/π bonding & Hückel $4n + 2$ torus
Paper VII	Materials Science	Streamtube network dimensionality (D)
Paper VIII	Quantum Materials	Global macroscopic phase locking ($\nabla\Phi = \text{const}$)
Paper IX	Self-Organization	Protein funnels & DNA double-helix topology
Paper X	Systems Biology	Metabolic strain reservoirs & neural phase pulses
Paper XI	Cognitive Dynamics	Predictive error minimization ($\min E_{\text{pred}}$)

- [2] R. Beem, *Substrate Molecular Self-Organization: Hydrogen Bonding, Protein Folding Topologies, DNA Helical Pairing, and Emergent Biological Structure*, UST Paper IX (2026).
- [3] R. Beem, *Substrate Quantum Materials: Global Phase Coherence, Cooper-Pair Topological Analogs, and Ab-Initio Superconductivity*, UST Paper VIII (2026).
- [4] R. Beem, *Substrate Materials: Carbon Allotropes, Graphene Transport, Fracture Mechanics, and Condensed Matter Structure*, UST Paper VII (2026).
- [5] R. Beem, *Substrate Chemistry II: Carbon Bonding Topologies, Topological Aromaticity, and Deterministic Reaction Dynamics*, UST Paper VI (2026).
- [6] R. Beem, *Substrate Chemistry I: Atomic Orbital Resonances, Topological Pauli Exclusion, Covalent Gradient Minimization, and Molecular Geometry*, UST Paper V (2026).
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- [8] R. Beem, *Substrate Electrodynamics: Fine-Structure Constant α and Precision Lepton Mass Hierarchy*, UST Paper III (2026).
- [9] R. Beem, *Ab-Initio Derivation of Proton Mass, Structure, and Charge Radius*, UST Paper II (2026).
- [10] R. Beem, *Topological Stabilization of 3D Solitons in Non-Linear Field Media*, UST Paper I (2026).